

ISUM — Input Sufficiency Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Basis Standard (PBAS)

Operational Layer: Prediction Basis Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Input Sufficiency

Version: 1.0

Status: Canonical · Open Module

Effective Date: 7 September 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Prediction Input Sufficiency

Canonical Definition

Input Sufficiency Module (ISUM) defines the governance conditions under which the identified, sourced, and relevance- assessed informational foundation of a prediction is evaluated for sufficient completeness, coverage, condition alignment, and material adequacy to support continuation of the Prediction Lifecycle.

ISUM determines whether the available governed Prediction Basis is sufficiently established for the applicable Prediction Object, Predictive Purpose, Prediction Target, Prediction Horizon, Predictive Boundary, Prediction Methodology, and known predictive conditions.

It distinguishes:



relevant information

from:

sufficient Prediction Basis.

A Prediction Basis may contain highly relevant information and still be insufficient.

Likewise, a Prediction Basis does not need to contain every possible piece of information in order to be sufficient.

ISUM therefore governs adequacy rather than informational maximalism.

More Information $\neq$ Sufficient Prediction Basis

Operational Role

ISUM serves as the fourth internal governance space of the Prediction Basis Standard (PBAS) .

It receives:

Identified Prediction Inputs

Governed Predictive Source Map

and:

Governed Relevant Signal Set

from the preceding PBAS modules.

The module asks:

Is the governed informational foundation sufficiently complete and materially adequate for this specific prediction to proceed under its stated methodology, purpose, horizon, boundaries, and conditions?

The governed progression is:

Identified Inputs + Source Structure + Relevant Signals $\rightarrow$ Sufficiency Criteria $\rightarrow$ Coverage Assessment $\rightarrow$ Material Gap Assessment $\rightarrow$ Sufficiency Determination $\rightarrow$ Governed Prediction Basis Sufficiency State

The result is handed to:

PBTRM — Prediction Basis Traceability Module.

Internal Module Flow

Prediction Input Identification $\rightarrow$ Predictive Source Mapping $\rightarrow$ Signal Relevance $\rightarrow \rightarrow$ Input Sufficiency $\rightarrow$ Prediction Basis Traceability

Current Position: Input Sufficiency

Module Operational Space

ISUM governs:

Prediction Basis sufficiency; informational completeness; material coverage; input coverage; source coverage; signal coverage; methodology-required inputs; target coverage; horizon coverage; boundary coverage; condition coverage; contextual coverage; temporal coverage; missing information; unavailable information; unresolved informational gaps; known unknowns; informational asymmetry; source concentration implications; signal concentration; dependency implications; conflict implications; redundancy implications; minimum sufficiency threshold; conditional sufficiency; restricted sufficiency; partial sufficiency; insufficiency; sufficiency uncertainty; sufficiency deterioration; reassessment; and historical sufficiency state preservation.

Module Function

ISUM prevents a prediction from progressing merely because some relevant information is available.

A prediction may possess:

valid inputs; credible sources; highly relevant signals; sophisticated methodology;

and still lack the informational foundation required to support its intended predictive claim.

For example:

an election forecast may omit entire regions; a failure prediction may lack critical operating-state data; a market forecast may rely upon one narrow source family; a medical prediction may lack required historical context; a short-term model may operate on stale information;

a collective prediction may lack meaningful informational diversity.

ISUM establishes the governance transition from:

Relevant Prediction Information

to:

Sufficiently Governed Prediction Basis

Prediction Basis Sufficiency

Prediction Basis Sufficiency is the governed state in which the informational foundation of a prediction is sufficiently complete, relevant, bounded, and materially adequate for the applicable predictive methodology and predictive frame.

Sufficiency is prediction-specific.

The same information may be sufficient for one prediction and insufficient for another.

Same Information Set $\neq$ Same Sufficiency State Across Predictions

Sufficiency Is Contextual

Sufficiency depends upon the governed predictive context.

It may vary according to:

Prediction Object; Predictive Purpose; Prediction Target; Prediction Horizon; Predictive Boundary; Prediction Methodology; information availability; uncertainty; operational conditions; system state; material dependencies.

ISUM therefore does not establish one universal quantity of information required for every prediction.

Sufficiency $\neq$ Completeness

A Prediction Basis does not need to contain all information that could theoretically be relevant.

The relevant governance question is whether missing information is materially significant.

Sufficiency $\neq$ Total Informational Completeness

A sufficiently governed Prediction Basis may still contain known gaps.

Those gaps must remain visible where material.

Methodology-Required Inputs

Prediction Methodology may require specific information before it can legitimately operate.

Examples may include:

defined variables;

minimum historical periods; specified sensor states; current market conditions; required contextual information; mandatory human assessments.

ISUM evaluates whether methodology-required informational conditions are satisfied.

Methodology Requirement Not Met $\neq$ Sufficient Prediction Basis by Default

Minimum Information Requirements

Where applicable, a methodology may define minimum informational conditions.

These may include:

minimum number of observations; minimum time range; required data classes; minimum geographical coverage; defined operating states; mandatory contextual fields.

ISUM may use these as part of the sufficiency determination.

It does not itself design the Prediction Methodology.

PMES retains methodology ownership.

Target Coverage

Prediction Basis must sufficiently cover the Prediction Target.

A dataset may contain extensive information while failing to represent the target population, system, event, or outcome structure.

Information Volume $\neq$ Target Coverage

Boundary Coverage

A Prediction Basis may be insufficient where information does not adequately represent the defined Predictive Boundary.

Examples include:

prediction covers five countries while inputs represent one; system forecast covers all factories while data represent only one plant; population forecast relies upon one demographic group.

Available Sample $\neq$ Sufficient Boundary Coverage

Prediction Horizon Coverage

Information sufficient for one horizon may not support another.

A short-term forecast may require current operational information.

A long-term forecast may require structural and historical information.

ISUM therefore evaluates sufficiency relative to the Prediction Horizon.

Short-Term Sufficiency $\neq$ Long-Term Sufficiency

Temporal Sufficiency

A Prediction Basis may be structurally complete but temporally inadequate.

Examples include:

stale market data; outdated system status; historical measurements used after regime change; delayed operational information.

Complete Historical Data $\neq$ Sufficient Current Basis

Condition Coverage

Where a prediction depends upon relevant operating, environmental, regulatory, market, or system conditions, the Prediction Basis must sufficiently represent those conditions.

Missing material condition information may create a Prediction Basis gap.

Contextual Coverage

Contextual information may be necessary even where it does not directly predict the target.

For example:

model outputs may be misleading without knowing operating regime; market signals may require liquidity context; weather predictions may require geographical context; failure predictions may require system configuration.

ISUM evaluates whether such contextual information is sufficiently represented.

Known Unknowns

A Known Unknown is a material informational element known to be unavailable, unresolved, uncertain, or incomplete at prediction time.

Known Unknowns do not automatically make Prediction Basis insufficient.

They must be assessed according to materiality.

Known Unknown $\neq$ Automatic Prediction Failure

Unknown Unknowns

ISUM cannot guarantee identification of information whose relevance or absence is genuinely unknown.

Therefore:

Prediction Basis Sufficiency $\neq$ Omniscience

Sufficiency reflects the governed informational state that can reasonably be established at the time.

Material Information Gaps

A Material Information Gap exists where absent, unavailable, incomplete, or unresolved information may materially impair the ability of Prediction Basis to support the prediction.

Material gaps should remain explicit.

They may lead to:

restriction; qualification; additional information request; reassessment; inability to proceed.

Missing Information

Missing information may arise because:

it does not exist; it cannot be accessed; it has not yet been generated; it was omitted; it was lost; its source cannot be established; it was excluded.

ISUM distinguishes the existence of a gap from the reason for the gap.

Unavailable Information

Information may be known to exist but unavailable.

Unavailable information does not automatically invalidate the Prediction Basis.

Its materiality must be assessed.

Unavailable Information $\neq$ Automatically Insufficient Prediction Basis

Excluded Information

A Prediction Basis may exclude information intentionally.

Where exclusion is material to sufficiency, ISUM requires that impact to be assessable.

The validity of the exclusion methodology may remain governed by PMES and relevant PBAS modules.

Source Coverage

Prediction Basis may require sufficient source coverage.

A prediction relying on many signals from one underlying source family may remain informationally narrow.

PSMM provides source dependency information.

ISUM determines whether that source structure creates a material sufficiency limitation.

Many Inputs from One Source Family $\neq$ Broad Source Coverage

Source Concentration Risk

High source concentration can create vulnerability where the source:

is incomplete; is biased; fails; changes; becomes unavailable; represents only part of the predictive environment.

High concentration does not automatically establish insufficiency.

It becomes a factor in sufficiency assessment.

Signal Coverage

Multiple relevant signals may represent the same underlying phenomenon.

ISUM therefore assesses whether the relevant signal set provides adequate informational coverage rather than merely a high signal count.

Signal Count $\neq$ Signal Coverage

Signal Redundancy

Redundancy can create false confidence.

Five relevant signals based on the same underlying information may provide less informational breadth than five independent signal families.

SRVM and PSMM establish the relevant relationships.

ISUM assesses the resulting sufficiency implications.

Conflicting Relevant Information

A Prediction Basis may contain conflicting relevant signals.

Conflict does not automatically make the basis insufficient.

In some cases, conflict is itself an important representation of uncertainty.

ISUM asks whether the methodology can legitimately proceed despite the conflict.

Information Conflict $\neq$ Automatic Insufficiency

Information Diversity

Information Diversity concerns the degree to which Prediction Basis reflects materially distinct informational perspectives, sources, variables, or mechanisms where diversity is relevant.

Diversity can strengthen sufficiency but is not always required.

Information Diversity $\neq$ Prediction Basis Sufficiency by Default

Informational Asymmetry

A predictor may possess materially different information from other predictors.

This does not itself make its Prediction Basis sufficient or insufficient.

ISUM evaluates the informational foundation on its own governed requirements.

PSMM retains the Predictor–Reality Proximity relationship.

Sufficiency Threshold

Where applicable, a Sufficiency Threshold defines the minimum governed informational state required for Prediction Basis to proceed.

The threshold may be established through:

Prediction Methodology; system governance requirements; domain-specific methodological rules; institutional controls; other legitimate governance criteria.

The threshold should not be invented retrospectively after the prediction outcome becomes known.

Sufficiency States

ISUM may establish the following states:

Sufficient

The Prediction Basis satisfies applicable informational requirements.

Conditionally Sufficient

The Prediction Basis is sufficient only while defined conditions or limitations hold.

Restricted

The Prediction Basis supports only a narrower predictive claim, purpose, boundary, or horizon than originally intended.

Partially Sufficient

Material informational support exists, but one or more requirements remain incomplete.

Insufficient

The Prediction Basis does not satisfy the minimum conditions necessary for legitimate continuation.

Unable to Determine

Available governance information does not permit a defensible sufficiency determination.

Conditional Sufficiency

A Prediction Basis may be sufficient only under stated conditions.

Examples:

only for a limited horizon; only within a defined geography;

only while current operating state persists; only where missing information remains non-material; only for exploratory rather than operational reliance.

Conditional sufficiency must not be presented as unrestricted sufficiency.

Restricted Sufficiency

A Prediction Basis may support a narrower prediction than originally proposed.

For example:

Nationwide Forecast → Regional Forecast

or:

12-Month Prediction → 30-Day Prediction

where available information legitimately supports only the narrower scope.

ISUM may therefore produce a restriction rather than binary rejection.

Partial Sufficiency

Partial Sufficiency identifies cases where Prediction Basis has meaningful support but does not fully satisfy the intended predictive claim.

Partial sufficiency should not automatically be represented as full sufficiency.

Insufficiency

A Prediction Basis is insufficient where material informational deficiencies prevent the prediction from legitimately proceeding under the stated predictive frame and methodology.

Insufficiency does not mean the predicted outcome will be wrong.

Insufficient Basis ≠ Incorrect Future Outcome

A poorly supported prediction may accidentally prove correct.

Sufficiency Uncertainty

The sufficiency determination itself may contain uncertainty.

Where uncertainty is material, ISUM should preserve it rather than forcing artificial precision.

Sufficiency Determination ≠ Certainty

Prediction Basis Gap

A Prediction Basis Gap is a material deficiency, unresolved condition, missing input, inadequate coverage, or informational limitation affecting Prediction Basis sufficiency.

Gaps may be:

informational; temporal; methodological;

contextual; source-related; target-related; boundary-related; condition-related.

Gap Materiality

Not every missing input constitutes a material gap.

ISUM assesses whether the gap could materially affect:

methodology operation; predictive interpretation; uncertainty; boundary legitimacy; horizon legitimacy; permitted reliance.

Gap Resolution

Possible responses include:

Additional Information

Clarification

Source Expansion

Signal Reassessment

Prediction Restriction

Methodology Reconsideration

Conditional Continuation

Inability to Proceed

ISUM identifies sufficiency consequences.

It does not automatically prescribe the broader organizational decision.

Prediction Basis Decay

A Prediction Basis initially determined sufficient may become less sufficient over time.

Potential causes include:

information becoming stale; material source loss; new contradictions; context change; regime change; target change; methodology change; operational changes.

Valid Initial Sufficiency $\neq$ Valid Continuing Sufficiency

Sufficiency Reassessment

Reassessment may be required where material conditions change.

Potential triggers include:

new information; lost information; changed source structure; signal relevance decay; changed prediction horizon; changed boundary; methodology version change; significant operational change.

Later PMRS governs Prediction Monitoring & Revision at the broader prediction level.

ISUM governs the informational sufficiency state.

Historical Sufficiency State

The sufficiency state applicable when the prediction was formed must remain reconstructable.

A prediction later supported by stronger information must not be retrospectively represented as having possessed that information originally.

Current Sufficiency $\neq$ Historical Sufficiency

Retrospective Sufficiency Inflation

After an outcome becomes known, later information may make the original prediction appear more justified than it actually was.

ISUM prevents retrospective enlargement of the original informational foundation.

Later Confirmation $\neq$ Original Prediction Basis Sufficiency

Input Sufficiency Determination

Input Sufficiency is sufficiently governed when:

applicable informational requirements are identifiable; material relevant inputs are available; required coverage is assessed; material source dependencies are considered; signal redundancy and conflict are

considered; known unknowns remain visible; material gaps are identified; sufficiency limitations are documented; conditional or restricted states remain explicit; sufficiency deterioration can trigger reassessment; historical sufficiency state remains reconstructable.

Governance Requirements

ISUM requires that:

1. sufficiency be assessed against the specific governed prediction; 2. Prediction Methodology requirements be considered; 3. target coverage be assessed; 4. Predictive Boundary coverage be assessed; 5. Prediction Horizon coverage be assessed; 6. temporal sufficiency be assessed where material; 7. contextual and condition coverage be considered; 8. known unknowns remain visible; 9. material missing information be identified; 10. unavailable information not automatically be treated as non-material; 11. source concentration implications be assessed where relevant; 12. signal redundancy not be mistaken for informational breadth; 13. conflicting relevant information remain visible; 14. information volume not be mistaken for sufficiency; 15. sufficiency thresholds be established before outcome knowledge where applicable; 16. conditional sufficiency remain distinguishable from unrestricted sufficiency; 17. restricted and partial sufficiency remain distinguishable from full sufficiency; 18. insufficiency not be equated with future prediction error; 19. material change trigger reassessment; 20. historical sufficiency state remain reconstructable.

Governance Logic

ISUM applies the following governance logic:

Identified Prediction Inputs + Governed Predictive Source Map + Governed Relevant Signal Set

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Applicable Sufficiency Requirements Identified

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Methodology-Required Inputs Assessed

↓

Target / Boundary / Horizon Coverage Assessed

↓

Temporal / Contextual / Condition Coverage Assessed

↓

Source Concentration / Dependency Implications Assessed

↓

Signal Redundancy / Conflict Implications Assessed

↓

Known Unknowns and Material Information Gaps Identified

↓

Gap Materiality Determined

↓

Sufficiency State Established

↓

Sufficient / Conditionally Sufficient / Restricted / Partially Sufficient / Insufficient / Unable to Determine

↓

Governed Prediction Basis Sufficiency State

↓

PBTRM — Prediction Basis Traceability Module

Where the Prediction Basis is materially insufficient:

Prediction Basis Governance Gap → Additional Information / Clarification / Restriction / Reassessment / Inability to Proceed

Minimum Implementation Framework

A minimum implementation of ISUM must establish:

A. Prediction Reference

The governed prediction or predictive frame whose Prediction Basis is being assessed.

B. Applicable Sufficiency Requirements

The informational conditions required by the predictive frame and Prediction Methodology.

C. Required Input Coverage

Material categories, inputs, conditions, or informational dimensions expected to be present.

D. Actual Input Coverage

The information actually represented in Prediction Basis.

E. Source Coverage

The material breadth, dependencies, and concentration of sources.

F. Signal Coverage

The materially distinct relevant signal structure.

G. Known Unknowns

Material information known to be unavailable, incomplete, unresolved, or uncertain.

H. Prediction Basis Gaps

Any missing or inadequate informational condition.

I. Gap Materiality

At minimum:

Non-Material / Potentially Material / Material / Critical / Unable to Determine

J. Sufficiency State

At minimum:

Sufficient / Conditionally Sufficient / Restricted / Partially Sufficient / Insufficient / Unable to Determine

K. Sufficiency Conditions

Any limitations under which sufficiency remains valid.

L. Reassessment Trigger

Conditions requiring a new sufficiency determination.

M. Historical Preservation

The sufficiency state and underlying basis applicable at prediction formation must remain reconstructable.

Governance Outputs

ISUM may produce:

Governed Prediction Basis Sufficiency State; Input Sufficiency Assessment; Sufficiency Requirements Record; required-input coverage record; actual-input coverage record; source-coverage profile; signal-coverage profile; Known Unknown Register; Prediction Basis Gap Register; gap-materiality determination; conditional-sufficiency record; restricted-sufficiency record; partial-sufficiency record; insufficiency determination; sufficiency uncertainty record; reassessment trigger; historical sufficiency record; unresolved information requirements.

The primary governed output is:

Governed Prediction Basis Sufficiency State

This becomes a principal input into:

PBTRM — Prediction Basis Traceability Module.

Critical Distinctions

Relevant Information $\neq$ Sufficient Prediction Basis

More Information $\neq$ Sufficient Prediction Basis

Sufficiency $\neq$ Total Informational Completeness

Same Information Set $\neq$ Same Sufficiency State Across Predictions

Methodology Requirement Not Met $\neq$ Sufficient Prediction Basis by Default

Information Volume $\neq$ Target Coverage

Available Sample $\neq$ Sufficient Boundary Coverage

Short-Term Sufficiency $\neq$ Long-Term Sufficiency

Complete Historical Data $\neq$ Sufficient Current Basis

Known Unknown $\neq$ Automatic Prediction Failure

Prediction Basis Sufficiency $\neq$ Omniscience

Unavailable Information $\neq$ Automatically Insufficient Prediction Basis

Many Inputs from One Source Family $\neq$ Broad Source Coverage

Signal Count $\neq$ Signal Coverage

Information Conflict $\neq$ Automatic Insufficiency

Information Diversity $\neq$ Prediction Basis Sufficiency by Default

Conditional Sufficiency $\neq$ Unrestricted Sufficiency

Partial Sufficiency $\neq$ Full Sufficiency

Insufficient Basis $\neq$ Incorrect Future Outcome

Sufficiency Determination $\neq$ Certainty

Valid Initial Sufficiency $\neq$ Valid Continuing Sufficiency

Current Sufficiency $\neq$ Historical Sufficiency

Later Confirmation $\neq$ Original Prediction Basis Sufficiency

Enterprise Application

ISUM enables organizations to answer a question that prediction systems frequently leave implicit:

Do we actually know enough to make this prediction in the form in which we intend to use it?

This matters across:

AI forecasting + financial prediction + industrial maintenance + supply-chain forecasting + insurance + healthcare + strategic planning + election forecasting + market prediction + operational decision systems.

An organization may possess sophisticated models and large datasets while still lacking critical informational coverage.

ISUM prevents predictive sophistication from masking informational insufficiency.

It creates a governed checkpoint before Prediction Basis proceeds toward final traceability and then into Prediction Assumptions.

Use Case 1 — A Nationwide Forecast Built from Data Covering Only Part of the Country

A logistics company develops an AI system to predict nationwide delivery demand.

The model processes millions of records and produces precise weekly forecasts.

The Prediction Inputs are well documented, the source structure is mapped, and the signals are individually relevant.

However, the historical data overwhelmingly represents large metropolitan areas.

Rural regions and several newly entered markets are sparsely represented.

The model nevertheless produces national forecasts without qualification.

The critical governance question is:

Can a Prediction Basis be considered sufficient for a nationwide prediction merely because it contains a large volume of relevant data when material parts of the defined Predictive Boundary are poorly represented?

ISUM evaluates informational coverage against the actual Predictive Boundary.

The result may be:

Restricted Sufficiency

for metropolitan markets,

rather than:

Full Sufficiency

for nationwide prediction.

Information Volume $\neq$ Sufficient Boundary Coverage

Use Case 2 — An Industrial Failure Prediction Has Strong Signals but Is Missing the One Operating State That Matters

An industrial company predicts the probability that a critical machine will fail within the next seven days.

The system has access to:

vibration data; temperature; pressure; maintenance history; historical failure records.

All are relevant.

However, a control-system interface has failed, and the predictive system no longer receives the machine's current operating-load state.

Historical analysis shows that identical vibration and temperature patterns have very different meanings depending on whether the machine is operating at 40% or 95% load.

The prediction system still produces a precise failure probability.

The critical governance question is:

Can the Prediction Basis remain sufficient when most relevant signals are present but a missing operational condition materially changes how those signals must be interpreted?

ISUM identifies the missing operating-load state as a potentially material Prediction Basis Gap.

The appropriate governance outcome may be:

Conditionally Sufficient

Restricted

or:

Insufficient

depending on the methodology and intended reliance.

Relevant Inputs $\neq$ Sufficient Prediction Basis

Canonical Boundary

ISUM governs the assessment of whether identified, sourced, and relevance-assessed Prediction Inputs collectively provide a sufficiently complete, materially adequate, appropriately bounded, and condition-aligned Prediction Basis for the governed prediction. It does not govern Prediction Input identification, detailed source mapping, Signal Relevance determination, evidence validity, predictive assumptions, Prediction Formation, probability representation, feasibility, Operational Reality, communication and reliance, or predictive performance.

ISUM establishes:

whether the informational foundation is sufficient for the prediction to legitimately proceed.

It does not establish:

whether the prediction will ultimately be correct.

Canonical Closing Statement

A prediction does not become governable because it has data, sources, or relevant signals. It becomes informationally governable when those elements are sufficient for the predictive claim being made.

ISUM establishes the point at which Prediction Basis becomes sufficiently complete to support legitimate continuation of the Prediction Lifecycle.

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